



Gaze-guided vision transformer for chest X-ray image classification

Zihui Chen, Zhi Liu^{*}, Yingjie Song

School of Communication and Information Engineering, Shanghai University, Shanghai 200444, China

ARTICLE INFO

Keywords:

Chest X-ray image
Image classification
Vision transformer
Gaze

ABSTRACT

Thoracic diseases pose significant public health challenges worldwide, with chest X-rays (CXR) playing a critical role in their diagnosis and monitoring. Recent advances in deep learning have enabled AI-driven diagnostic systems to enhance the accuracy and efficiency of CXR classification. However, the interpretability of these models remains a key concern. We propose a novel approach that integrates expert gaze data into a Vision Transformer (ViT) to improve interpretability in CXR classification. Our method integrates gaze information through the Gaze Information Injector (GII), which directs the model's attention to clinically relevant regions based on expert knowledge. To reduce overfitting, the GII module compresses gaze features while aligning the model's focus with expert gaze patterns. To ensure robustness, our training strategy includes both gaze-annotated and non-annotated samples, allowing inference without requiring gaze input. We evaluate our model on the EG-CXR and OMSI datasets, showing improved classification performance compared to state-of-the-art models. Ablation studies validate the effectiveness of the GII module, while visualization results highlight improved interpretability by focusing on disease-related regions in CXR images.

1. Introduction

Thoracic diseases have become one of the major public health issues worldwide, leading to significant mortality and morbidity. Chest X-ray (CXR) is a common radiographic imaging technique that plays a crucial role in the diagnosis and monitoring of thoracic diseases. CXR images can clearly display lung structures and potential abnormalities, including pneumonia, tuberculosis, lung cancer, and chronic obstructive pulmonary disease (COPD). Studies have shown that early diagnosis and treatment can significantly improve patient prognosis and survival rates [1]. However, manual analysis of image is time-consuming and labor-intensive, and it relies on the expertise and experience of radiologists. This dependence introduces subjectivity and increases the risk of misdiagnosis [2].

With the advancement of deep learning technology, AI-based automated diagnostic systems in medical image analysis has garnered widespread attention, particularly in CXR image classification. Automated diagnostic systems not only enhance diagnostic efficiency but also assist experts in making more accurate decisions, thereby improving patient outcomes and quality of life [3]. Despite their success, a critical challenge remains: the interpretability of deep learning models.

Deep learning models are often seen as “black boxes”, making it difficult to understand their internal decision-making processes. This lack of transparency reduces the credibility of these models in practical medical applications, especially when life-critical diagnoses are

involved. Studies have shown that deep learning models sometimes make decisions based on regions unrelated to the disease, leading to erroneous diagnostic results [4]. For example, models might be influenced by irrelevant features in CXR images, such as text markers or shadows of other organs, resulting in incorrect judgments [5].

To address this issue, some researchers have attempted to incorporate expert gaze data into models. In the field of medical imaging, radiologists and other healthcare professionals analyze and interpret images by selectively focusing on specific regions based on their extensive experience and knowledge, which are crucial for identifying pathologies [6]. By introducing expert gaze data, models can learn more meaningful features, thereby improving their decision accuracy and interpretability [7]. Expert gaze data can provide valuable information about the regions of interest in CXR images, guiding the model to focus on the most relevant regions for diagnosis. This scheme can help alleviate the negative effect of irrelevant features and enhance the model's interpretability by aligning its attention with the expert's gaze patterns.

Previous works combining expert gaze data primarily employed attention consistency methods, using gaze data as a supervisory signal. Auxiliary tasks were designed to supervise the attention maps of models, employing multi-task learning where the model learns both the main and auxiliary tasks. The features learned from the auxiliary task enhance the performance of the main task. While multitask learning effectively extracts domain-specific information from gaze data

^{*} Corresponding author.

E-mail address: liuzhisjtu@163.com (Z. Liu).

<https://doi.org/10.1016/j.bspc.2025.108298>

Received 10 October 2024; Received in revised form 2 April 2025; Accepted 27 June 2025

Available online 22 July 2025

1746-8094/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

to improve classification performance, the supervisory signal is not particularly stable and contains uncertainties. In practice, the dataset samples gaze data from a single physician, whose attention regions can be highly variable. For instance, the physician may focus on non-disease-related areas of CXR images, such as embedded devices. These specific instances significantly increase the complexity of the auxiliary task, leading to overfitting and inhibiting the training of the main task.

To overcome these challenges, we propose an alternative approach that treats gaze data as an additional input rather than as a supervisory signal. Our model processes both CXR images and gaze data during training, but randomly omits gaze input for a subset of samples. This strategy ensures that the model can make predictions even in the absence of gaze data. Without treating gaze data as a supervisory signal, we can design modules within our model to flexibly process gaze information.

Specifically, we design a Gaze Information Injector (GII) module to fuse gaze information within a Vision Transformer (ViT). The GII module is positioned placed between the output of the Multi-Head Self Attention (MHSA) and the output of the Feed-Forward Network (FFN) of the transformer block. It adjusts image features using gaze features to enhance interpretability. In this module, we downscale gaze features using a Multi-Layer Perceptron (MLP) to reduce overfitting risks, then restore them to original dimension. Simultaneously, we downscale the output features of MHSA to reduce information unrelated to attention weights, then adjust these weights using spatial attention mechanism with features extracted from expert's gaze points.

From the above, the main contributions of our work are summarized as follows:

- (1) We propose a Gaze Information Injector (GII) module to integrate expert gaze information into ViT for CXR image classification. The GII module adopts a bottleneck structure for refining feature representations to facilitate transfer learning. Additionally, it employs a Gaze Feature Fusion (GFF) block with spatial attention to merge gaze and image features effectively.
- (2) Unlike previous methods that use gaze data for auxiliary supervision or preprocessing, we treat gaze as an input modality. This approach allows the modules to extract and integrate expert domain knowledge flexibly while mitigating overfitting associated with additional supervision. To ensure robust inference, we randomly exclude gaze data for a subset of training samples, allowing the model to generalize to cases without gaze annotations.
- (3) We validated our Gaze Information Injector based Vision Transformer (GII-ViT) on two relevant datasets. To demonstrate the model's effectiveness, we conducted ablation studies and compared it with state-of-the-art gaze-guided models. Experimental results indicate that our proposed model achieves improved performance, setting new benchmarks in the field.

2. Related works

2.1. Studies with gaze

Gaze contains rich information about human intentions, and studying gaze points can provide insights into human cognitive processes and behavior analysis. Several studies have attempted to incorporate human attention into AI models through gaze points, impacting fields such as visual recognition [8], emotion recognition [9], visual question answering [10,11], image enhancement [12], and autonomous driving [13, 14]. Additionally, some methods [15–18] have successfully predicted eye gaze points in natural images, achieving high performance.

In the medical field, several studies focus on the gaze points of experts while interpreting medical images. Experts' gaze point contain valuable domain-specific information, and analyzing these points can provide deeper insights into the visual diagnostic process [19,20]. Some studies [21,22] have shown that using expert's eye positions as visual aids can enhance the diagnostic performance of novice readers. These findings lay the theoretical foundation for integrating gaze information into computer-aided diagnostic systems, leveraging expert gaze data to improve system accuracy and reliability.

2.2. Chest X-ray image classification

Deep learning models have achieved remarkable performance in CXR image classification tasks. Rajpurkar et al. proposed CheXNet [3], a deep learning model based on DenseNet [23], which achieved state-of-the-art performance on the ChestX-ray14 dataset. CheXNet was trained on 112,120 frontal-view CXR images and could detect 14 common thoracic diseases. Wang et al. introduced a multi-label classification model called CheXpert [24], which used a 121-layer DenseNet to classify 14 thoracic diseases. CheXpert was trained on 224,316 CXR images and achieved better performance than CheXNet. Jiang et al. [25] proposed a Transformer-based dual-path decoder that utilizes learnable label embeddings and effectively connects local discriminative features with corresponding labels for improved thoracic disease classification. Öztürk et al. [26] integrate a transformer backbone with a multi-branch output module, effectively emphasizing critical regions while maintaining sensitivity to the co-occurrence relationships among various diseases. However, these models face challenges in interpretability, making it difficult to understand their decision-making processes.

2.3. Gaze-guided models

Attention Consistency Models: These models only require chest X-ray images as input, with expert gaze points used as supervision. The specific process involves calculating the model's attention map using CAM (Class Activation Map) or gaze prediction methods, and then supervising the model-generated attention map with expert gaze points to enhance the model's attention alignment with the expert's gaze patterns. Typically, these models adopt a multi-task learning approach, where both classification loss and attention loss are used to supervise the model during training. Karargyris et al. [7] proposed a benchmark model based on U-Net [27] using the EG-CXR dataset, which contains 1083 chest X-ray images with expert gaze data. Their model used a U-shaped network for gaze prediction and added a classification head at the bottom of the encoder part, training both parts simultaneously. Zhu et al. [28] noted that in multi-task learning, there might be competition between attention consistency learning and classification learning, potentially leading to overfitting. They introduced an uncertainty-based [29] multi-task balancing scheme to improve the model's robustness. Unlike the gaze prediction-based methods mentioned above, Zhu H et al. [30] proposed a more direct way to regulate model attention by starting with a classification model and using the CAM method to generate the model's attention map, thereby reducing the model's complexity by not requiring additional parameters for gaze prediction.

Inspired by these methods, some researchers have introduced similar techniques into medical images beyond chest X-rays. Wang et al. [31] applied this method to the task of osteoarthritis grading and proposed a bone joint expert gaze dataset. They considered the phenomenon of random gaze wandering, which could introduce uncertainty into the gaze points. To address this issue, they used a conditional Gaussian probability model to model the uncertainty of gaze points. They also compared results under different training dataset sizes and found that the effect of injecting expert gaze points was more significant with smaller datasets. Jiang et al. [32] introduced this method to the task of diabetic retinopathy detection and created a lightweight gaze dataset. They observed that when experts gaze at healthy images with no abnormalities, their behavior is more random, and the uncertainty of gaze points is higher. In contrast, when viewing unhealthy images containing abnormal regions, the gaze tends to focus on suspicious regions, resulting in more stable gaze points. These points are more likely to include abnormalities and are more valuable for disease classification. Therefore, they proposed class-aware factors to adjust the attention loss weights for different categories of images.

Due to the inherent variability in expert attention, training the model to achieve strong alignment with expert attention through supervisory signals is challenging and may lead to overfitting. Therefore,

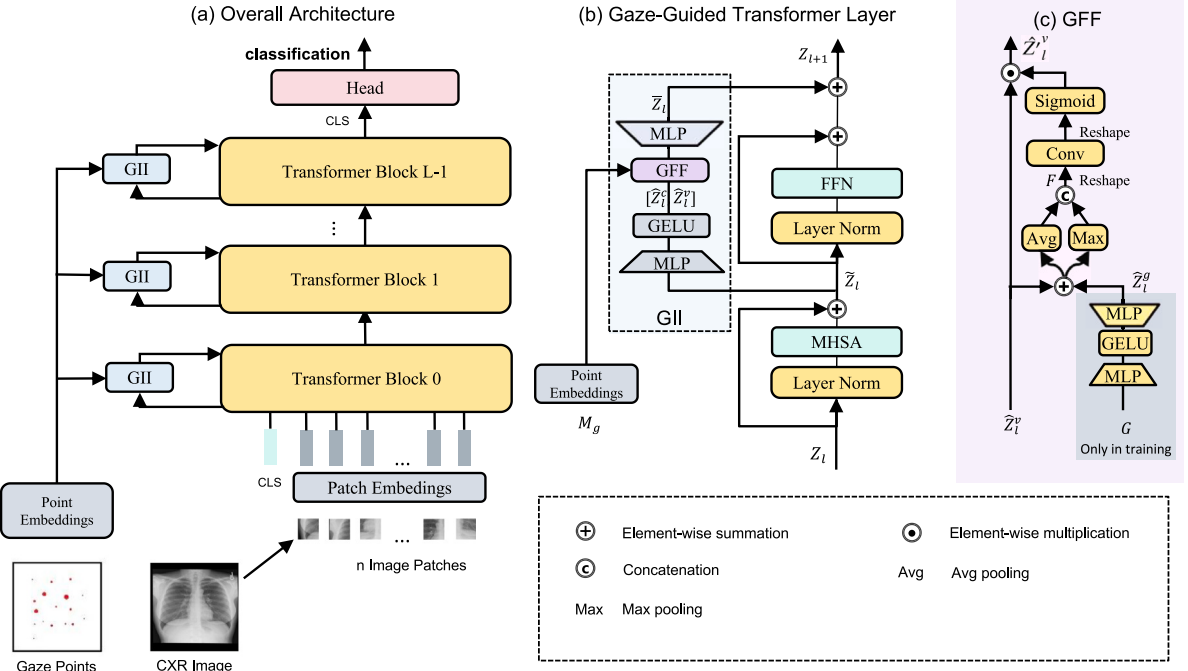


Fig. 1. Overall architecture of the proposed model.

instead of using gaze points as a supervisory signal, we incorporate them as an input source. We design modules to extract expert gaze information and inject it into the model. Through our training strategy, we ensure that during inference, our model only requires the image as input.

Preprocessing models: Unlike the attention consistency models that use expert gaze points as a supervisory signal, these models use expert gaze points for preprocessing images. Rong et al. [33] used generated gaze density maps to set thresholds and occlude parts of the image below the threshold. They fed the filtered image and the original image into different encoder networks to extract information, then integrated the outputs to get classification results. Ma et al. [34] referred to the MAE [35] training method, replacing the original random selection of image patches to be discarded with targeted selection based on gaze points. Using ViT [36] as the backbone network, they divided the image into patches and ranked each patch's importance based on the number of gaze points. They then discarded patches of lower importance proportionally, allowing the network to focus on the remaining patches related to the disease, preventing the model from overfitting irrelevant regions. Additionally, this model requires gaze input only during training for the discard operation, with only the image needed during inference.

The above methods force the model to make classification decision based on the potentially disease-related regions highlighted by experts. They effectively use expert's gaze to filter out irrelevant regions. However, reducing expert gaze features to a binary map leads to information loss and might not fully leverage the expert's prior knowledge. In our work, we embed the gaze points into the network and gradually integrate the gaze information through a well-designed GII module, which helps the model handle semantically rich scenes.

3. The proposed model

3.1. Architecture overview

The overall framework of the proposed GII-ViT is illustrated in Fig. 1(a), which consists a stack of gaze-guided transformer layers and a classification head. It processes CXR images with expert gaze

embeddings. In gaze-guided Transformer layers, our Gaze Information Injector (GII) module integrates expert knowledge into the features extracted from CXR images. Disease classification is conducted using a global average pooling layer followed by a fully connected layer in the classification head. The training strategy involves selectively excluding gaze data during iterations to facilitate inference without gaze points.

3.2. Gaze embedding

Gaze data is collected by an eye-tracking device at a certain sampling rate, consisting of a series of scatter points. Each scatter point represents the position where the expert focuses on the image. These scatter points not only contain positional information but also gaze duration information, representing the expert's attention to the area. In the lower left corner of Fig. 1(a), there is a gaze map with many red dots of varying sizes, where the size of each red dot indicates the duration of the expert's gaze in that area.

We can represent the collected gaze points with a set $[(x_1, y_1, t_1), (x_2, y_2, t_2), \dots, (x_n, y_n, t_n)]$, where (x_i, y_i) is the position of the i th gaze point and t_i is the gaze duration. We use a map $M_g \in \mathbb{R}^{H \times W}$ to represent these gaze points. The value at each spatial position in M_g corresponds to the duration of the expert's gaze at that point, while regions not gazed upon are initialized with zero. M_g is usually a very sparse map with only a few non-zero points. To enable the gaze map M_g to interact with the image features in the transformer, it is then normalized to the range $[0, 1]$ and embedded into a fixed-dimensional vector $G \in \mathbb{R}^{n \times d}$ using a linear projection.

3.3. Gaze-guided transformer layer

3.3.1. Transformer block

In this section, we describe the transformer block utilized in our GII-ViT. We begin by detailing the input processing stage. Initially, patch embedding is performed on the input CXR image. Following the methodology of ViT, the input CXR image I is divided into a grid of non-overlapping patches, each of size $P \times P$. These patches are subsequently flattened and linearly projected into a fixed-dimensional space, resulting in patch embeddings. For an image comprising n patches,

where $n = \frac{H \times W}{p^2}$, the sequence of patch embeddings $\mathbf{Z}_0 \in \mathbb{R}^{(n+1) \times d}$ is formulated as:

$$\mathbf{Z}_0 = [\text{CLS}; \mathbf{E}p_1, \mathbf{E}p_2, \dots, \mathbf{E}p_n] + \mathbf{E}_{\text{pos}} \quad (1)$$

where CLS denotes the classification token, p_i represents the flattened i th patch of image, $\mathbf{E}$ is the linear projection matrix, and $\mathbf{E}_{\text{pos}}$ represents the positional encodings.

At the l th layer, the transformer block processes the input sequence $\mathbf{Z}_l$ using Multi-Head Self Attention (MHSA) and Feed-Forward Network (FFN). The operations for the l th layer can be described as follows:

$$\tilde{\mathbf{Z}}_l = \text{MHSA}(\text{LN}(\mathbf{Z}_l)) + \mathbf{Z}_l \quad (2)$$

$$\mathbf{Z}_{l+1} = \text{FFN}(\text{LN}(\tilde{\mathbf{Z}}_l)) + \tilde{\mathbf{Z}}_l + \tilde{\mathbf{Z}}_l \quad (3)$$

where LN denotes Layer Normalization, $\tilde{\mathbf{Z}}_l$ is the output from the GII module, which will be described in detail in the following subsection. The residual connections are employed to facilitate better gradient flow and mitigate the vanishing gradient problem during training.

This block-wise transformation continues until the final transformer block is reached, ensuring that the output embeddings progressively capture more complex and abstract features necessary for the CXR image classification task.

3.3.2. Gaze information injector

In this subsection, we propose the proposed GII module, which integrates gaze information into the model via an additional input module, enabling feature extraction aligned with expert knowledge. The GII module consists of a bottleneck architecture and a GFF block, connected to the MHSA output of each transformer block via skip connections.

Due to the small scale of the CXR image dataset, it is insufficient to train the ViT from scratch, and it is necessary to inherit the pre-trained ViT parameters. However, unlike natural images, CXR images exhibit significant variations in the scale and texture of lesions, resulting in different sample-specific features, which can complicate transfer learning. To address this, inspired by [37–39], we propose a bottleneck architecture into GII module enhance adaptation to CXR images. This design facilitates effective fine-tuning when transferring a ViT model pre-trained on natural images to CXR classification tasks. The bottleneck structure refines feature representations for the target task while preserving knowledge from pre-training, thereby improving transfer learning efficiency. Additionally, it enhances semantic feature extraction from CXR images, contributing to model robustness and performance, and its effectiveness is validated in Section 4.4.

Specifically, given the input $\tilde{\mathbf{Z}}_l \in \mathbb{R}^{(n+1) \times d}$, the output feature vector of multi-head self-attention in the l th gaze-guided transformer layer, we apply a compression linear layer to reduce its dimensionality, which can be expressed as follows:

$$\hat{\mathbf{Z}}_l = \text{GELU}(\text{MLP}_{\text{down}}(\tilde{\mathbf{Z}}_l)) \quad l = 0, 1, 2, \dots, L-1 \quad (4)$$

where MLP_{down} represents a compression layer with weights $d \times d'$, GELU represents the activation function, and $\hat{\mathbf{Z}}_l \in \mathbb{R}^{(n+1) \times d'}$ represents the downscaled feature.

To incorporate expert gaze information, we propose the Gaze Feature Fusion (GFF) block to integrate expert gaze information into image features. The GFF block uses spatial attention to guide the model's focus toward disease-relevant regions, mimicking how radiologists examine CXR images. The details of the GFF block are illustrated in Fig. 1(c).

Before passing features to the GFF block, we *split* $\hat{\mathbf{Z}}_l \in \mathbb{R}^{(n+1) \times d'}$ into an image feature token $\hat{\mathbf{Z}}_l^v \in \mathbb{R}^{n \times d'}$ and a class token $\hat{\mathbf{Z}}_l^c \in \mathbb{R}^{1 \times d'}$, which are formulated as follows:

$$[\hat{\mathbf{Z}}_l^c, \hat{\mathbf{Z}}_l^v] = \text{split}(\hat{\mathbf{Z}}_l) \quad (5)$$

The GFF block takes the embedded gaze token $\mathbf{G} \in \mathbb{R}^{n \times d}$ and the compressed image feature token $\hat{\mathbf{Z}}_l^v \in \mathbb{R}^{n \times d'}$ as inputs. To mitigate

potential overfitting caused by redundant gaze information cause by the uncertainty of expert gaze points, we perform feature compression to retain classification-relevant information and reduce redundant information that could lead to overfitting. Specifically, gaze features are projected into a low-rank representation via $\text{MLP}_{\text{down}} \in \mathbb{R}^{d \times d_g}$, followed by transformation to match the dimensionality of $\hat{\mathbf{Z}}_l^v$ using GELU and $\text{MLP}_{\text{tr}} \in \mathbb{R}^{d_g \times d'}$. This process is formulated as follows:

$$\hat{\mathbf{Z}}_l^g = \text{MLP}_{\text{tr}}(\text{GELU}(\text{MLP}_{\text{down}}(\mathbf{G}))) \quad (6)$$

where $\hat{\mathbf{Z}}_l^g \in \mathbb{R}^{n \times d'}$ represents the downscaled gaze token.

Subsequently, we utilize spatial attention to fuse image features with gaze features, enhancing the model's focus on disease-related regions. Specifically, we stack the downscaled gaze features and image features together, perform average pooling and max pooling respectively, and concatenate them into the feature $\mathbf{F}$, which is formulated as follows:

$$\mathbf{F} = \text{Concat} \left[\text{Avg} \left(p\hat{\mathbf{Z}}_l^g + \hat{\mathbf{Z}}_l^v \right), \text{Max} \left(p\hat{\mathbf{Z}}_l^g + \hat{\mathbf{Z}}_l^v \right) \right] \quad (7)$$

where the parameter p is related to our training strategy, with a value of either 0 or 1, to control whether gaze points are utilized.

Next, We reshape $\mathbf{F}$ to a 2D feature map and apply convolutional layers to capture local dependencies between adjacent regions before restoring it to its original dimension. The sigmoid function generates spatial attention weights, which are then multiplied with $\hat{\mathbf{Z}}_l^v$ to obtain the adjusted image feature $\hat{\mathbf{Z}}_l'^v$.

$$\hat{\mathbf{Z}}_l'^v = \hat{\mathbf{Z}}_l^v \odot \sigma \left(rs^{-1}(\text{Conv}(rs(\mathbf{F}))) \right) \quad (8)$$

where rs represents reshaping the feature into a 2D feature map, and rs^{-1} represents restoring the 2D feature map back to token form, σ represents sigmoid operation, $\odot$ represents the element-wise multiplication operation.

Next, The output feature $\hat{\mathbf{Z}}_l'^v$ is then concatenated with the class token $\hat{\mathbf{Z}}_l^c$ and processed by the upsampling layer MLP_{up} to match the transformer's feature dimension, yielding the final output of the GII module, $\tilde{\mathbf{Z}}_l \in \mathbb{R}^{(n+1) \times d}$, as follows:

$$\tilde{\mathbf{Z}}_l = \text{MLP}_{\text{up}} \left(\text{Concat} \left(\hat{\mathbf{Z}}_l^c, \hat{\mathbf{Z}}_l'^v \right) \right) \quad (9)$$

Finally, we integrate the transformed features with transformer block's FFN output via a residual connection, as shown in Eq. (3).

3.4. Training strategy

Considering that in real-world scenarios, the gaze points of experts are not accessible during the inference phase, we adapt our training process accordingly. In each training batch, we randomly select 50% of CXR image samples to be processed without GII, which equals to the case that no gaze points are used as input, while the remaining samples are processed with GII.

4. Experiments

4.1. Datasets

Our study primarily focuses on the task of chest X-ray image classification. In publicly available datasets, We used the two datasets for our experiments: EG-CXR and OMSI. Both datasets consist of images drawn from existing large-scale chest X-ray image collections, annotated with gaze points by medical experts.

EG-CXR [40]: Introduced by Karargyris and colleagues, this dataset is derived from the MIMIC-CXR [41,42] database and includes a total of 1,083 CXR images. Each image is accompanied by aligned data including the transcribed radiological report text, radiologist's dictation audio, and eye gaze coordinates. The dataset encompasses three categories: pneumonia (360 images), congestive heart failure (CHF, 363 images), and normal (360 images), with balanced age and gender

Table 1

Performance comparison of gaze-guided models on the EG-CXR and OMSI datasets. The results are averaged over 10 runs with 95% confidence intervals.

(a) EG-CXR [40]					
Method	AUC	Specificity	Sensitivity	Precision	F1-Score
MTL-Unet (2022) [28]	0.887 ± 0.017	0.835 ± 0.015	0.782 ± 0.012	0.786 ± 0.013	0.779 ± 0.007
CG-CAM (2021) [30]	0.888 ± 0.015	0.843 ± 0.017	0.770 ± 0.019	0.770 ± 0.016	0.770 ± 0.011
Gaze-MTL (2021) [43]	0.900 ± 0.008	0.856 ± 0.015	0.775 ± 0.014	0.780 ± 0.014	0.776 ± 0.007
EG-ViT (2023) [34]	0.909 ± 0.013	0.873 ± 0.011	0.807 ± 0.011	0.807 ± 0.012	0.807 ± 0.008
Ours	0.919 ± 0.011	0.891 ± 0.010	0.808 ± 0.012	0.804 ± 0.013	0.806 ± 0.007

(b) OMSI [43]					
Method	AUC	Specificity	Sensitivity	Precision	F1-Score
MTL-Unet (2022) [28]	0.832 ± 0.011	0.763 ± 0.021	0.748 ± 0.014	0.735 ± 0.015	0.741 ± 0.008
CG-CAM (2021) [30]	0.830 ± 0.014	0.769 ± 0.017	0.745 ± 0.015	0.729 ± 0.019	0.737 ± 0.010
Gaze-MTL (2021) [43]	0.833 ± 0.016	0.775 ± 0.018	0.752 ± 0.012	0.723 ± 0.013	0.737 ± 0.007
EG-ViT (2023) [34]	0.859 ± 0.013	0.791 ± 0.012	0.761 ± 0.011	0.751 ± 0.016	0.756 ± 0.009
Ours	0.867 ± 0.012	0.814 ± 0.012	0.773 ± 0.013	0.754 ± 0.012	0.763 ± 0.007

distribution within each category. In the absence of an official dataset partition, we adopted a 9:1 split for our training and testing sets.

OMSI [43]: Selected from the SIIM-ACR [44] pneumothorax dataset, it comprises 5,777 images, forming a binary classification dataset with 78% normal and 22% pneumothorax samples. Out of these, 1,250 images were reviewed by radiologists, and gaze data was collected. To ensure fairness in comparison, we adopted the official dataset division method, use 1,000 images for training and 250 images for testing.

4.2. Experimental setup

In this study, our model was implemented using the PyTorch framework, with enhancements through data augmentation techniques. The augmentations included random resizing and cropping of CXR and gaze maps to 224×224 , random horizontal flipping, and normalization. The backbone network utilized was the pre-trained ViT-B/16 model, which underwent contrastive learning on approximately 400 million image-text pairs. The Adam optimizer was employed with an initial learning rate of 0.0001 and a weight decay of 0.0001. Cross-entropy was used as the loss function for classification. All training was performed on an NVIDIA Titan XP graphics card with 12 GB RAM. During the testing phase, images were resized to 224×224 pixels and normalized in the same manner as in the training phase.

To address the class imbalance in the OMSI dataset (78% normal vs. 22% pneumothorax), we employed the focal loss [45] instead of standard cross-entropy loss. This helps mitigate the bias toward the majority class by down-weighting the normal class samples, allowing the model to focus more on the minority pneumothorax cases. the focal loss is defined as:

$$L_{\text{focal}}(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t), \quad (10)$$

where p_t is the predicted probability for the true class, α is a weighting factor for class imbalance, and γ is a focusing parameter that reduces the loss contribution from well-classified examples, directing the model's attention to hard-to-classify cases. The hyperparameters were set to $\gamma = 2$ and $\alpha = 0.78$.

4.3. Comparative studies

In this section, we evaluate our model on the EG-CXR and OMSI datasets. We use the following five metrics to evaluate our model: the area under the receiver operating characteristic curve (AUC), Sensitivity, Specificity, Precision, and F1-score. To ensure statistical reliability, all results are averaged over 10 runs with 95% confidence intervals.

We compare our model with state-of-the-art gaze-guided models, including MTL-Unet [28], CG-CAM [30], Gaze-MTL [43], and EG-ViT [7]. For a fair comparison, all models were trained under identical settings.

Table 1 presents the comparative results on both datasets. Our model consistently outperforms baseline methods across all metrics on

Table 2

Metrics for patient subgroups in the EG-CXR dataset. The results are averaged over 10 runs with 95% confidence intervals.

Category	AUC	Specificity	Sensitivity	Precision	F1-Score
Normal	0.919 ± 0.011	0.904 ± 0.015	0.809 ± 0.012	0.805 ± 0.013	0.807 ± 0.007
CHF	0.931 ± 0.012	0.916 ± 0.017	0.819 ± 0.012	0.807 ± 0.015	0.813 ± 0.008
Pneumonia	0.909 ± 0.012	0.877 ± 0.015	0.795 ± 0.014	0.800 ± 0.014	0.797 ± 0.007

the OMSI dataset. On the EG-CXR dataset, our model achieves the highest AUC and Sensitivity, highlighting its effectiveness in correctly identifying pathology-related cases. However, its Precision and F1-score are slightly lower than those of EG-ViT, suggesting a trade-off between detecting positive cases and minimizing false positives.

Furthermore, we further analyze the performance on the EG-CXR dataset for each class. Table 2 presents the classification metrics for each category. Across different disease categories, our model achieves relatively strong performance, particularly showing effectiveness in detecting CHF (AUC = 0.931). This suggests that our approach captures certain pathological patterns associated with CHF. However, we observe variability in sensitivity and specificity across classes, likely reflecting inherent differences in disease manifestation. For example, pneumonia cases exhibit lower specificity, which may be attributed to overlapping radiographic features with other respiratory conditions.

Additionally, Fig. 2 illustrates the receiver operating characteristic (ROC) curves of the model with performance closest to the average across 10 runs for both the EG-CXR and OMSI datasets. These curves provide insights into the model's discriminative ability across varying classification thresholds. ROC curves plot the true positive rate (TPR) against the false positive rate (FPR) at various classification thresholds. Curves that are closer to the top-left corner and exhibit a steeper trend indicate better classification performance. The area under the curve (AUC) serves as a quantitative indicator, with higher values reflecting stronger discriminative capability.

4.4. Ablation studies

We conducted extensive ablation studies to evaluate how various design choices affect the classification performance. The factors evaluated include the dimensionality of gaze features and output features of MHSA, and the role of various components.

In Table 3, we verify the importance of each part of the proposed model. The results indicate that the introduction of GII and GFF significantly improves the classification performance of our model.

Dimensionality Comparison: we discuss the determination process of two hyperparameters: the bottleneck architecture compression dimension d' in GII and the compression dimension d_g in GFF. Initially, we removed the GFF block from the GII module and conducted transfer learning using only CXR images. Experimental results show

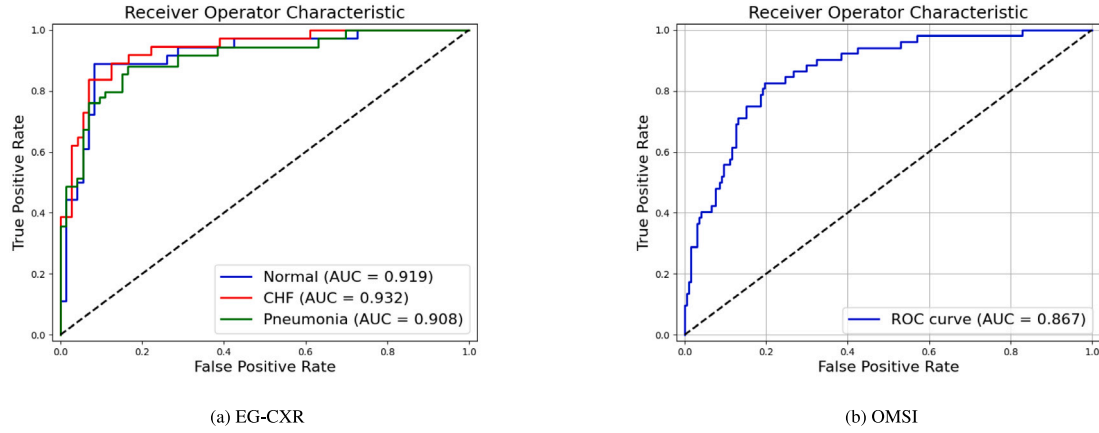


Fig. 2. Receiver Operating Characteristic (ROC) curves and corresponding AUC scores of our method on the EG-CXR and OMSI datasets. The curves represent models with performance closest to the average across 10 runs.

Table 3

Ablation studies of our method on EG-CXR dataset with ViT-B/16 backbone.

Settings	AUC	Specificity	Sensitivity	Precision	F1-Score
ViT	0.893	0.841	0.752	0.724	0.737
+GII w/o GFF	0.899	0.866	0.754	0.760	0.757
+GFF	0.906	0.881	0.787	0.788	0.787
Ours	0.919	0.891	0.808	0.804	0.806

Table 4

Ablation studies of dimension compression on EG-CXR dataset with ViT-B/16 backbone.

(a) Ablation study of dimension compression in bottleneck architecture.

Num of d'	AUC	F1-Score
10	0.903	0.783
20	0.919	0.806
30	0.912	0.801
40	0.909	0.795
50	0.904	0.791

(b) Ablation study of dimension compression in GFF.

Num of d_g	AUC	F1-Score
10	0.908	0.785
20	0.915	0.802
30	0.919	0.806
40	0.917	0.798
50	0.913	0.792

that the optimal compression dimension for transfer learning is $d' = 20$. Subsequently, we fixed d' at 20, reintroduced the GFF block, and employed our training strategy to identify $d_g = 30$ as the most suitable compression dimension.

Finally, to validate these choices, we conducted further experiments by fixing one hyperparameter while varying the other. The results, as shown in Table 4(a) and Table 4(b), confirm that $d' = 20$ and $d_g = 30$ are the optimal dimensions. Table 4(a) demonstrates that when $d' = 20$, both AUC and F1-Score achieve their peak values. Similarly, Table 4(b) illustrates that fixing d' at 20 and adjusting d_g reveals 30 as the optimal value, achieving the highest performance.

4.5. Visualization

To better understand the impact of our model design and analyze its interpretability, we use Grad-CAM [46] for visualization. The generated attention maps help determine whether the model focuses on the regions related to chest diseases in the images. Grad-CAM uses gradients to calculate the model's attention map, requiring no modifications to the model structure, making it easily deployable in the ViT model.

Fig. 3 presents visualization results for some images in the EG-CXR dataset. The first row shows the original CXR images with disease-related regions marked by red dashed circles. The second row shows attention maps generated by directly fine-tuning the original ViT model, while the third row shows attention maps generated using the proposed model with expert gaze points. It can be observed that, in the absence of expert domain knowledge, the original ViT model may base its classification decisions on irrelevant regions, such as the edges in the second row. However, when using the proposed model that incorporates radiologists' visual attention, the model focuses more on disease-related regions. Thus, from these examples, we can conclude that the decisions made by our ViT model are more interpretable, as they are guided by the radiologists' gaze points to focus on disease-related regions.

5. Critical discussion

5.1. Real-world applicability: Annotation efficiency, interpretability, and extensibility

The primary contribution of this study is the introduction of the Gaze Information Injector (GII) module, which integrates expert gaze data into the Vision Transformer (ViT) framework. Unlike previous methods that use gaze data solely as a supervisory signal, our approach innovatively treats gaze information as an additional input source, processed through a specialized module placed between the multi-head self-attention and feed forward network components of the transformer block.

One of the key advantages of our approach is its potential to enhance model interpretability. The use of gaze-based attention mechanisms allows for more transparent decision-making by highlighting the features most relevant to predictions. This level of transparency is critical for clinical acceptance, as it builds trust among healthcare professionals and facilitates the integration of AI technology into routine diagnostic workflows.

Furthermore, compared to traditional annotation methods, which typically require medical experts to perform detailed pixel-wise annotations of lesion regions across numerous images, gaze data collection is relatively less intrusive and can be seamlessly integrated into the daily workflow of radiologists. This reduces the annotation burden while still capturing expert domain knowledge, making the approach more practical for real-world deployment.

In addition, the proposed model introduces the GII module without altering the core architecture of Vision Transformer. This modular design ensures compatibility with standard Vision Transformer models and offers strong extensibility, allowing for the integration of additional input modalities, such as textual reports or electronic health records, through similarly structured injection modules. Such flexibility facilitates future adaptations of the framework to broader multimodal clinical applications.

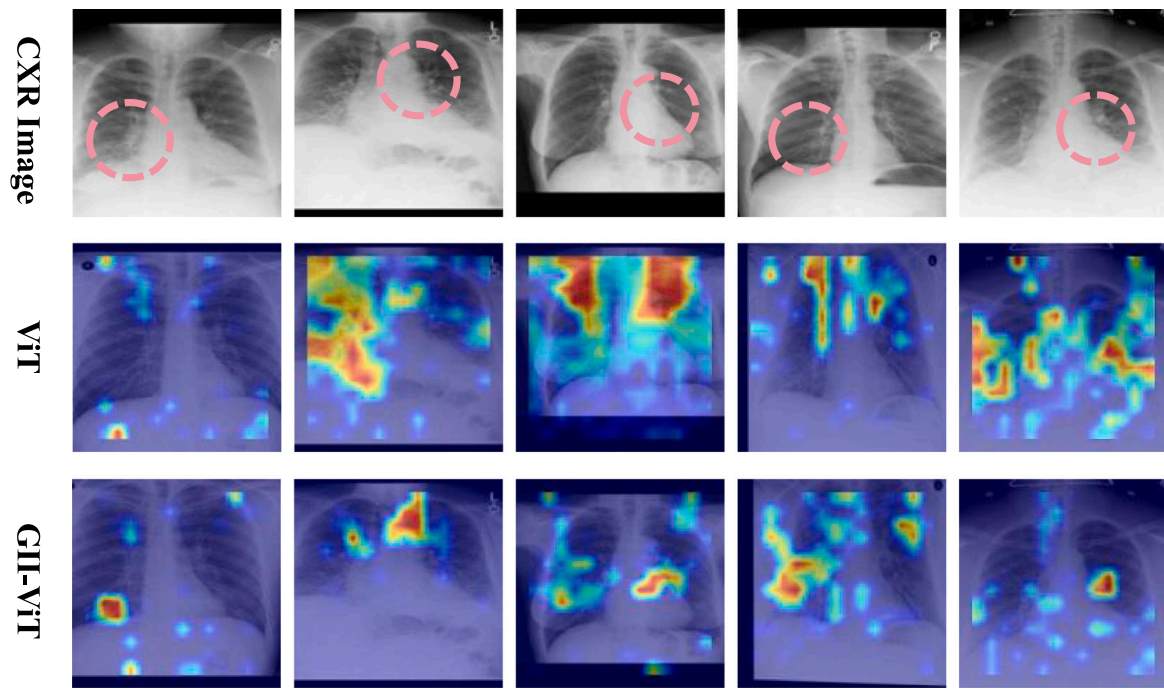


Fig. 3. The first row shows the original CXR images, the second row shows the attention maps obtained with ViT, and the third row shows the attention maps obtained with our GII-ViT.

5.2. Challenges in clinical workflow integration

Despite these advantages, several challenges remain in integrating our approach into clinical practice. One major limitation is the reliance on the EG-CXR and OMSI datasets, which contain gaze data from a single expert. In real-world applications, multiple radiologists may have varying gaze patterns due to differences in experience, diagnostic strategies, or personal biases. The variability in gaze distributions among experts could pose a challenge for large-scale deployment, as the model may overfit to a specific observer's pattern rather than generalizing across multiple clinicians. It would require strategies such as multi-expert gaze aggregation, personalized adaptation techniques, or uncertainty modeling to handle inter-expert variability effectively.

Additionally, while our model enhances interpretability by incorporating gaze information, its generalization ability has not yet been fully validated in real-world clinical settings, requiring further validation to assess its robustness and effectiveness across different clinical environments.

5.3. Future directions

To improve the scalability and robustness of our approach, future work could focus on extending the model to incorporate gaze data from multiple experts to better capture diagnostic variability and reduce bias introduced by single-observer supervision. Potential solutions include aggregation methods or uncertainty modeling to handle inter-observer variability more effectively. Another promising direction is the integration of additional modalities, such as radiology reports or structured clinical data to provide a more comprehensive understanding of disease patterns. In addition, validation across diverse clinical settings and patient populations will be necessary to ensure robustness and generalizability. By exploring these avenues, gaze-based AI models could play a more significant role in advancing interpretable and clinically applicable deep learning methods in medical image.

6. Conclusion

In this paper, we proposed a novel framework for integrating expert gaze data into ViT for CXR image classification. The main contribution is the GII module, which directly incorporates gaze information to guide attention toward disease-relevant regions. By leveraging feature compression techniques and a bottleneck design, our method mitigates overfitting risks and enhances transfer learning. Unlike prior approaches that use gaze data as a supervisory signal, our model treats it as an additional input, enabling inference solely on CXR images. Experimental results on two datasets demonstrate superior classification performance and improved interpretability compared to state-of-the-art gaze-guided models.

There are also some limitations in our study. Our model relies on gaze data from a single expert, which may introduce individual-specific biases, potentially limiting generalizability. We plan to explore integrating multi-expert gaze data using aggregation methods or uncertainty modeling to address inter-observer variability in future. Additionally, further validation in diverse clinical settings is necessary to confirm the model's robustness and effectiveness in realworld scenarios. Future research will also extend the framework by integrating additional modalities, such as radiology reports or structured clinical data, to provide a more comprehensive understanding of disease patterns.

CRedit authorship contribution statement

Zihui Chen: Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Zhi Liu:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Yingjie Song:** Writing – review & editing, Visualization, Validation, Formal analysis.

Dataset and ethical considerations

This study utilizes publicly available chest X-ray datasets, which have been de-identified and released for research purposes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant 62171269.

Data availability

Data will be made available on request.

References

- [1] Sanjeet Bagchi, WHO's global tuberculosis report 2022, *Lancet Microbe* 4 (1) (2023) e20.
- [2] Geert Litjens, Thijs Kooi, Babak Ehteshami Bejnordi, Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Mohsen Ghafoorian, Jeroen AWM Van Der Laak, Bram Van Ginneken, Clara I Sánchez, A survey on deep learning in medical image analysis, *Med. Image Anal.* 42 (2017) 60–88.
- [3] Pranav Rajpurkar, Jeremy Irvin, Kaylie Zhu, Brandon Yang, Hershel Mehta, Tony Duan, Daisy Ding, Aarti Bagul, Curtis Langlotz, Katie Shpanskaya, et al., Chexnet: Radiologist-level pneumonia detection on chest x-rays with deep learning, 2017, arXiv preprint arXiv:1711.05225.
- [4] John R Zech, Marcus A Badgeley, Manway Liu, Anthony B Costa, Joseph J Titano, Eric Karl Oermann, Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study, *PLoS Med.* 15 (11) (2018) e1002683.
- [5] Marcus A Badgeley, John R Zech, Luke Oakden-Rayner, Benjamin S Glicksberg, Manway Liu, William Gale, Michael V McConnell, Bethany Percha, Thomas M Snyder, Joel T Dudley, Deep learning predicts hip fracture using confounding patient and healthcare variables, *NPJ Digit. Med.* 2 (1) (2019) 31.
- [6] A Van der Gijp, CJ Ravesloot, H Jarodzka, MF Van der Schaaf, IC Van der Schaaf, Jan PJ van Schaik, Th J Ten Cate, How visual search relates to visual diagnostic performance: a narrative systematic review of eye-tracking research in radiology, *Advances in Health Sciences Education* 22 (3) (2017) 765–787.
- [7] Alexandros Karargyris, Satyananda Kashyap, Ismi Lourentzou, Joy T Wu, Arjun Sharma, Matthew Tong, Shafiq Abedin, David Beymer, Vandana Mukherjee, Elizabeth A Krupinski, et al., Creation and validation of a chest X-ray dataset with eye-tracking and report dictation for AI development, *Sci. Data* 8 (1) (2021) 92.
- [8] Kyle Min, Jason J. Corso, Integrating human gaze into attention for egocentric activity recognition, in: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2021, pp. 1069–1078.
- [9] Joochan Kim, Jungryul Seo, Teemu H. Laine, Detecting boredom from eye gaze and EEG, *Biomed. Signal Process. Control.* 46 (2018) 302–313.
- [10] Ekta Sood, Fabian Kögel, Philipp Müller, Dominike Thomas, Mihai Băce, Andreas Bulling, Multimodal integration of human-like attention in visual question answering, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 2648–2658.
- [11] Shi Chen, Ming Jiang, Jinhui Yang, Qi Zhao, Air: Attention with reasoning capability, in: *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part I* 16, Springer, 2020, pp. 91–107.
- [12] S Mahdi H Miangoleh, Zoya Bylinskii, Eric Kee, Eli Shechtman, Yağiz Aksoy, Realistic saliency guided image enhancement, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 186–194.
- [13] Jianwu Fang, Dingxin Yan, Jiahuan Qiao, Jianru Xue, Hongkai Yu, DADA: Driver attention prediction in driving accident scenarios, *IEEE Trans. Intell. Transp. Syst.* 23 (6) (2021) 4959–4971.
- [14] Jianwu Fang, Dingxin Yan, Jiahuan Qiao, Jianru Xue, He Wang, Sen Li, DADA-2000: Can driving accident be predicted by driver attention f analyzed by a benchmark, in: *2019 IEEE Intelligent Transportation Systems Conference, ITSC, IEEE*, 2019, pp. 4303–4309.
- [15] Yingjie Song, Zhi Liu, Gongyang Li, Dan Zeng, Tianhong Zhang, Lihua Xu, Jijun Wang, RINet: Relative importance-aware network for fixation prediction, *IEEE Trans. Multimed.* (2023).
- [16] Ziqiang Wang, Zhi Liu, Weijie Wei, Huizhan Duan, SalED: Saliency prediction with a pithy encoder-decoder architecture sensing local and global information, *Image Vis. Comput.* 109 (2021) 104149.
- [17] Xinhui Luo, Zhi Liu, Weijie Wei, Linwei Ye, Tianhong Zhang, Lihua Xu, Jijun Wang, Few-shot personalized saliency prediction using meta-learning, *Image Vis. Comput.* 124 (2022) 104491.
- [18] Jiawei Xie, Zhi Liu, Gongyang Li, Yingjie Song, Audio-visual saliency prediction with multisensory perception and integration, *Image Vis. Comput.* 143 (2024) 104955.
- [19] Harold L. Kundel, Paul S. La Follette Jr., Visual search patterns and experience with radiological images, *Radiology* 103 (3) (1972) 523–528.
- [20] Trafton Drew, Melissa L.-H. Võ, Jeremy M. Wolfe, The invisible gorilla strikes again: Sustained inattention blindness in expert observers, *Psychol. Sci.* 24 (9) (2013) 1848–1853.
- [21] Damien Litchfield, Linden J Ball, Tim Donovan, David J Manning, Trevor Crawford, Viewing another person's eye movements improves identification of pulmonary nodules in chest x-ray inspection., *J. Exp. Psychol.: Appl.* 16 (3) (2010) 251.
- [22] Ellen M Kok, Anique BH De Bruin, Simon GF Robben, Jeroen JG Van Merriënboer, Looking in the same manner but seeing it differently: Bottom-up and expertise effects in radiology, *Appl. Cogn. Psychol.* 26 (6) (2012) 854–862.
- [23] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, Kilian Q Weinberger, Densely connected convolutional networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 4700–4708.
- [24] Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silvana Ciurea-Ilcus, Chris Chute, Henrik Marklund, Behzad Haghighi, Robyn Ball, Katie Shpanskaya, et al., Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, 33, (01) 2019, pp. 590–597.
- [25] Xiaoben Jiang, Yu Zhu, Yatong Liu, Gan Cai, Hao Fang, TransDD: A transformer-based dual-path decoder for improving the performance of thoracic diseases classification using chest X-ray, *Biomed. Signal Process. Control.* 91 (2024) 105937.
- [26] Şaban Öztürk, M. Yiğit Turalı, Tolga Çukur, HydraViT: Adaptive multi-branch transformer for multi-label disease classification from chest X-ray images, *Biomed. Signal Process. Control.* 100 (2025) 106959.
- [27] Olaf Ronneberger, Philipp Fischer, Thomas Brox, U-net: Convolutional networks for biomedical image segmentation, in: *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015: 18th International Conference, Munich, Germany, October 5–9, 2015, Proceedings, Part III* 18, Springer, 2015, pp. 234–241.
- [28] Hongzhi Zhu, Robert Rohling, Septimiu Salcudean, Jointly boosting saliency prediction and disease classification on chest X-ray images with multi-task unet, in: *Annual Conference on Medical Image Understanding and Analysis*, Springer, 2022, pp. 594–608.
- [29] Alex Kendall, Yarin Gal, Roberto Cipolla, Multi-task learning using uncertainty to weigh losses for scene geometry and semantics, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7482–7491.
- [30] Hongzhi Zhu, Septimiu Salcudean, Robert Rohling, Gaze-guided class activation mapping: Leverage human visual attention for network attention in chest x-rays classification, in: *Proceedings of the 15th International Symposium on Visual Information Communication and Interaction*, 2022, pp. 1–8.
- [31] Sheng Wang, Xi Ouyang, Tianming Liu, Qian Wang, Dinggang Shen, Follow my eye: Using gaze to supervise computer-aided diagnosis, *IEEE Trans. Med. Imaging* 41 (7) (2022) 1688–1698.
- [32] Hongyang Jiang, Yilin Hou, Hanpei Miao, Haili Ye, Mengdi Gao, Xiaoling Li, Richu Jin, Jiang Liu, Eye tracking based deep learning analysis for the early detection of diabetic retinopathy: a pilot study, *Biomed. Signal Process. Control.* 84 (2023) 104830.
- [33] Yao Rong, Wenjia Xu, Zeynep Akata, Enkelejda Kasneci, Human attention in fine-grained classification, 2021, arXiv preprint arXiv:2111.01628.
- [34] Chong Ma, Lin Zhao, Yuzhong Chen, Sheng Wang, Lei Guo, Tuo Zhang, Dinggang Shen, Xi Jiang, Tianming Liu, Eye-gaze-guided vision transformer for rectifying shortcut learning, *IEEE Trans. Med. Imaging* (2023).
- [35] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, Ross Girshick, Masked autoencoders are scalable vision learners, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 16000–16009.
- [36] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xi-aohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, 2020, arXiv preprint arXiv:2010.11929.
- [37] Yaoming Zhu, Jiangtao Feng, Chengqi Zhao, Mingxuan Wang, Lei Li, Counter-interference adapter for multilingual machine translation, in: *Findings of the Association for Computational Linguistics: EMNLP 2021*, 2021, pp. 2812–2823.
- [38] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, Lora: Low-rank adaptation of large language models, 2021, arXiv preprint arXiv:2106.09685.
- [39] Kaidong Zhang, Dong Liu, Customized segment anything model for medical image segmentation, 2023, arXiv preprint arXiv:2304.13785.
- [40] Alexandros Karargyris, Satyananda Kashyap, Ismi Lourentzou, Joy Wu, Matthew Tong, Arjun Sharma, Shafiq Abedin, David Beymer, Vandana Mukherjee, Elizabeth Krupinski, et al., Eye gaze data for chest X-rays (version 1.0.0), 2020, 2020, URL <https://physionet.org/content/egd-cxr/1.0.0>.

- [41] Alistair EW Johnson, Tom J Pollard, Seth J Berkowitz, Nathaniel R Greenbaum, Matthew P Lungren, Chih-ying Deng, Roger G Mark, Steven Horng, MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports, *Sci. Data* 6 (1) (2019) 317.
- [42] Alistair EW Johnson, Tom J Pollard, Nathaniel R Greenbaum, Matthew P Lungren, Chih-ying Deng, Yifan Peng, Zhiyong Lu, Roger G Mark, Seth J Berkowitz, Steven Horng, MIMIC-CXR-JPG, a large publicly available database of labeled chest radiographs, 2019, arXiv preprint [arXiv:1901.07042](https://arxiv.org/abs/1901.07042).
- [43] Khaled Saab, Sarah M Hooper, Nimit S Sohoni, Jupinder Parmar, Brian Pogatchnik, Sen Wu, Jared A Dunnmon, Hongyang R Zhang, Daniel Rubin, Christopher Ré, Observational supervision for medical image classification using gaze data, in: *Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part II* 24, Springer, 2021, pp. 603–614.
- [44] Anna Zawacki, Carol Wu, George Shih, Julia Elliott, Mikhail Fomitchev, Mohannad Hussain, Paras Lakhani, Phil Culliton, Shunxing Bao, SIIM-ACR Pneumothorax Segmentation, Kaggle, 2019, URL <https://kaggle.com/competitions/siim-acr-pneumothorax-segmentation>.
- [45] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, Piotr Dollár, Focal loss for dense object detection, *IEEE Trans. Pattern Anal. Mach. Intell.* 42 (2) (2020) 318–327, <http://dx.doi.org/10.1109/TPAMI.2018.2858826>.
- [46] Ramprasaath R Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, Dhruv Batra, Grad-cam: Visual explanations from deep networks via gradient-based localization, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 618–626.